

Comprehensive Analysis of Recent DDoS Attacks (2022–2025)

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Introduction

Over the past few years, Distributed Denial of Service (DDoS) attacks have become one of the most disruptive cyber threats, capable of crippling even the world’s most secure infrastructures. Between 2022 and 2025, there has been a sharp escalation in both the scale and sophistication of such attacks, with threat actors exploiting advanced protocols, leveraging massive botnets, and even weaponizing legitimate cloud resources.

This report analyzes **five major DDoS attacks** from the last three years, summarizing their scale, methods, and effects. Following that, one particularly significant attack — the **Google Cloud DDoS attack of 2023** — is investigated in depth, focusing on the target, technology used, motives, impact, and mitigation strategies.

Five Major DDoS Attacks (2022–2025)

Year	Target	Type of Attack	Peak Traffic	Description
2025	Cloudflare	HTTP/2 “Rapid Reset” Exploit	201 million RPS	Exploited a flaw in HTTP/2 protocol allowing attackers to flood servers with fake requests.
2024	Microsoft (Outlook, OneDrive, Azure)	Layer 7 HTTP Flood	70 million RPS	Aimed at disrupting user access to core Microsoft online services globally.
2023	Google Cloud Platform	HTTPS Flood via HTTP/2 Exploit	398 million RPS	The largest DDoS attack ever recorded, overwhelming Google’s front-end servers.
2023	Ukrainian Government Websites	Network-Level Flooding	N/A	Geopolitically motivated attack targeting national online infrastructure during conflict.

Year	Target	Type of Attack	Peak Traffic	Description
2022	Wynn Resorts	IoT Botnet Amplification	20 million RPS	Conducted using a botnet of compromised smart devices and routers.

Each of these attacks highlights how DDoS threats have evolved — from simple network floods to intelligent, adaptive assaults exploiting vulnerabilities in application-layer protocols like HTTP/2 and HTTP/3.

Detailed Investigation: Google Cloud DDoS Attack (August 2023)

1. Target

The **Google Cloud Platform (GCP)** was the primary target. Specifically, the attack focused on the **HTTP/S Load Balancer** infrastructure responsible for managing inbound client traffic to Google-hosted applications and websites. The intention behind targeting Google Cloud was strategic — a successful disruption could have cascaded across hundreds of dependent services and client organizations that rely on GCP for hosting, authentication, and networking.

Google’s cloud infrastructure is globally distributed, handling millions of legitimate HTTPS requests per second from users and businesses. The attackers attempted to saturate this system with an unprecedented volume of forged traffic, aiming to exhaust both computational resources and bandwidth.

The attack vector primarily focused on:

- Overwhelming the **front-end edge servers** responsible for SSL/TLS termination.
- Disrupting **load balancing algorithms**, thereby causing service degradation across multiple regions.
- Targeting **shared infrastructure**, meaning even unaffected clients could experience latency or denial of access due to collateral load.

This choice of target demonstrated deep reconnaissance and understanding of cloud-level infrastructure dependencies.

2. Technology Used

The attackers employed an **Application Layer DDoS (Layer 7)** strategy, focusing on **HTTP/2 protocol vulnerabilities**, particularly a mechanism later referred to as the **“Rapid Reset” exploit**.

This vulnerability allowed an attacker to open multiple streams within a single HTTP/2 connection and immediately cancel them, forcing the server to allocate resources repeatedly without completing any legitimate request. By automating this process at scale, attackers were able to amplify their attack without requiring proportional bandwidth.

Technical Details:

- **Attack Type:** HTTPS Flood via HTTP/2 Rapid Reset Exploit
- **Peak Intensity:** 398 million requests per second (RPS), the largest ever recorded
- **Source:** A globally distributed **botnet** composed of compromised virtual private servers (VPS), routers, and IoT devices
- **Duration:** Approximately 3–5 minutes of peak load, sustained enough to test Google’s upper defense limits
- **Tools & Frameworks Suspected:** Custom DDoS scripts leveraging automated orchestration over compromised cloud APIs

Mechanism:

1. The botnet opened thousands of concurrent HTTP/2 streams.
2. Before each stream could transmit data, the attacker immediately issued a “RST_STREAM” (reset) frame.
3. The server still processed these as legitimate requests, allocating memory and CPU for each one.
4. Repeating this cycle at extreme speed led to massive **resource exhaustion**, equivalent to billions of incomplete handshakes per second.

This method allowed attackers to use **fewer devices but generate exponentially higher load**, showing a shift from traditional brute-force DDoS to *protocol exploitation*.

3. Attacker’s Motive

While the identities of the attackers remain unconfirmed, several possible motives were identified by Google's threat intelligence and external researchers:

- **Demonstration of Capability:** Cybercriminal groups often conduct large-scale attacks to showcase the power of their botnets, attracting clients on the dark web offering "DDoS-for-hire" services.
- **Testing Corporate Defenses:** Attacks on major tech firms like Google are sometimes used to benchmark defensive technologies before targeting less-protected enterprises.
- **Disruption for Political or Economic Impact:** Some analysts believe the attack was a politically motivated act designed to disrupt Western cloud services that support global digital infrastructure.
- **Ransom or Extortion Potential:** Although no ransom demand was publicly reported, DDoS extortion campaigns are a known tactic where attackers threaten to continue attacks unless payments are made.

Overall, the attack was likely **motivated by a combination of prestige, testing, and psychological impact**, aiming to demonstrate that even the most robust cloud providers are not invulnerable.

4. Overall Impact

Despite the scale, **Google successfully mitigated the attack**, making this incident a success story for defensive resilience.

However, its **global cybersecurity impact** was significant:

- **Technical Impact:**
The attack consumed enormous network capacity and processing power. Google's defensive systems automatically scaled to absorb the flood by diverting traffic across its global backbone.
- **Operational Impact:**
No direct service outages were reported, but the event temporarily stressed multiple regional data centers, leading to **slight latency spikes** in Asia-Pacific and North America.
- **Industry Impact:**
The event triggered immediate global awareness among major cloud and hosting

providers. Vendors such as Amazon, Microsoft, and Cloudflare issued **emergency advisories and patches** to their HTTP/2 implementations.

- **Reputational Impact:**

Although Google's mitigation was effective, the attack raised concern about the **future sustainability** of cloud-based DDoS protection when faced with near half-billion RPS events.

The 2023 Google Cloud incident thus became a **benchmark case study** in cyber defense, illustrating how rapid-response automation and distributed mitigation can prevent service failure even under record-breaking stress.

5. Defensive Strategies

In response to the attack, Google outlined both **proactive** and **reactive** defense mechanisms that proved effective and recommended for future mitigation:

a. Protocol Hardening:

The core vulnerability was within HTTP/2. Google collaborated with major web server vendors (e.g., Apache, Nginx) and browser developers to introduce stricter handling of stream resets. Protocol updates were later rolled out to block repeated rapid-reset sequences.

b. Adaptive Traffic Filtering:

Using AI-based anomaly detection, Google Cloud Armor dynamically identified abnormal traffic patterns and differentiated legitimate sessions from malicious ones. This minimized false positives while cutting off attack sources almost instantly.

c. Global Load Distribution:

Traffic was rerouted through Google's **Anycast-based global network**, absorbing surges across multiple regions rather than concentrating load at single data centers.

d. Layered Defense Approach:

Google implemented a **multi-layered mitigation pipeline** — from edge routers filtering suspicious IP blocks, to application firewalls inspecting request headers and connection behavior.

e. Zero Trust Architecture:

Sensitive backend services were isolated from the public internet, ensuring that only validated edge nodes interacted with core systems, preventing attackers from pivoting deeper.

f. Industry Collaboration:

Google shared telemetry and findings with the broader security community, leading to **CVE assignments** and cross-industry protocol updates that prevented similar large-scale attacks in the future.

Conclusion

The **2023 Google Cloud DDoS attack** redefined the limits of what a denial-of-service assault could achieve, both in scale and complexity. While Google's systems proved resilient, the attack revealed the fragility of modern internet infrastructure against protocol-level exploits.

The incident emphasizes the following takeaways:

- **Continuous Evolution of Threats:** Attackers now exploit protocol logic, not just bandwidth.
- **Need for AI-Powered Defense:** Automated, intelligent response mechanisms are essential at global scale.
- **Collaborative Security:** Transparency and shared mitigation data between companies and cybersecurity communities are crucial.

Ultimately, the attack served as both a **warning and a learning opportunity**, pushing the cybersecurity world to fortify its foundations against a new era of high-speed, protocol-based DDoS warfare.
